

Meeting Preparation: Proposal Discussion

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What you are defending, in one sentence

A fixed-effects panel study of roughly 28 Indian states over roughly 25 years, testing whether satellite-measured PM2.5 rises and then falls as state income grows, with the National Clean Air Programme tested as an intervention.

Questions he is likely to ask, and strong answers

“Where is the mathematics in this?”

Expect this one. It is a mathematics project, not an economics project, and this is the most likely challenge. The honest answer is that the mathematical content sits in the inference, not the application: panel unit-root asymptotics, the distinction between first- and second-generation tests under cross-sectional dependence, and above all Wagner's point that polynomial transformations of integrated processes have non-standard limiting behaviour, so the usual cointegration machinery does not apply. That last point is a genuine limit-theory issue and is the mathematical spine of the project. If he wants more, the cointegrating polynomial regression extension is where you would go.

“Explain Wagner's critique in your own words.”

If income is $I(1)$, income squared is not $I(1)$; it is not integrated of any order. So running unit-root and cointegration tests and then treating the quadratic EKC regression as a cointegrating relationship is not valid, and much of the published EKC literature does exactly that. Your response is to make no cointegration claim at all: estimate within-state associations using fixed effects with Driscoll-Kraay standard errors, and say plainly that this is an association, not a long-run equilibrium relationship.

“Why satellite data instead of the official CPCB monitors?”

Because the ground network expanded substantially over the study period and is concentrated in metropolitan areas, with much of rural India unmonitored. A state's measured pollution therefore changes when monitors are added, which confounds measurement change with real change, precisely the kind of bias that would manufacture a spurious trend. Satellite estimates give complete, consistent coverage every year, and extend the panel to about 25 years instead of 10. You retain CPCB data as a validation check.

“What if you find nothing?”

A null result is a real result here, and worth saying so confidently. The literature is already split. One recent paper finds an inverted-N shape, another finds policy matters more than income, so evidence that the simple inverted-U does not hold for Indian states is a genuine contribution, not a failed project.

“Have you actually got the data yet?”

Be honest about where you are. If you have not downloaded anything, say so and commit to having the satellite files and one state's GSDP series in hand before the next meeting. Do not claim more than is true; supervisors notice.

“Is this feasible in the time you have?”

Yes, because nothing here requires model training or new methodology. It is data assembly plus standard panel estimation. The heaviest step is processing gridded satellite files into state-level aggregates, which is a GIS-style task rather than a research risk.

Weak points: concede these rather than defend them

- Causality. You cannot separate income driving pollution from pollution dragging on income. Say so before he does, and describe the results as associational.
- Turning point. If the estimated turning point lies outside the observed income range, it is an extrapolation, not a prediction. Report it as such.
- Spillovers. Punjab and Haryana stubble burning shows up in Delhi's air, so state-level attribution is imperfect. Spatial models are the honest 'future work' answer.
- Population denominator. The 2021 Census never happened, so post-2011 per-capita figures rest on projections.

One gap to raise yourself

The proposal says satellite PM2.5 will be aggregated to states using population weighting, but it does not name a source for gridded population, which requires a separate dataset such as NASA SEDAC's Gridded Population of the World. Raise this yourself rather than letting him find it: it shows you have thought the pipeline through to the implementation level. The alternative is simple area-weighted averaging, which is easier but less meaningful, since it treats empty desert the same as a dense city.

Questions to ask him

- Is a fixed-effects panel with no cointegration claim an acceptable response to Wagner for a project at this level, or would he want the cointegrating polynomial regression attempted?
- Population-weighted or area-weighted aggregation of the satellite grid: does he have a preference?
- How much of the mark rests on methodological rigour versus the novelty of the result?
- What does he want to see at the next meeting, and how often should you check in?
- Is there a preferred format or length for the final report?

Before you walk in

Reread Sections 4 and 5 of your own proposal so the data sources and the five methodology steps are fresh. Take notes in the meeting, and send a short follow-up afterwards confirming what was agreed.